

TERRAIN DATABASE

FORMAT DESCRIPTION DOCUMENT

(TDBFDD)

	Name	Function	Date and Signature
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Ref: J43055AC 01

Product: TDB	Classif.: Proprietary reserve	Date: 17.04.2025
Program: GCAM	Category: Type approval submission	Language: English
		Number of pages: 37

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Entity Identifier	Document Identifier	Document Type Code	Revision
0026-F9111	J43055AC	506	01

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LIST OF REVISIONS

Issue	DATE	Description of revisions	Author
J43055AA 00A	30/08/04	Preliminary issue built from AIS reference 28.10.66.11 revision 1.5	N. MARTY
J43055AA 00	26/05/05	First issue after peer revue AMS/05/002985 IR 00	D. BARBEY
J43055AB 00A	31/03/10	Preliminary issue for introduction of low Rnp (T3CAS)	D. BARBEY
J43055AB 00	14/04/10	Release after peedr review : AMS/10/002512.	D. BARBEY
J43055AB 01	25/02/25	Add format precision on segment name field	M. Girardet
J43055AC 00	20/03/25	Change output delivery format (from CD to web server or hard drive)	M. Girardet
J43055AC 01	17/04/25	Take into account internal review for DPAL1 terrain database	M. Girardet

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1 Purpose And Organisation

This document describes the data and file format of the internal Thales digital elevation models (DEM) as provided for example for the TAWS database production.

This document is divided into the following sections:

- Section 1 gives a short introduction in the purpose and the organization of the document.
- Section 2 describes general properties of the terrain database.
- Section 3 describes the detailed organization of the database, like file system structure, naming conventions, and file organization.
- Sections 4 and 5 describe all data elements in detail, e.g. their types, contents and sizes.

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2 Introduction

The Thales Terrain Database offers a grid-based digital elevation model (DEM) covering the whole world in a 30" resolution and selected areas in 15" or 3" resolution . Each grid cell represents a (nearly) rectangular 30" by 30" (or 15" by 15" or 3" by 3") area of the world and holds specific data about this area. In the project on hand, the following data are provided:

Maximum elevation (ELV):

The ELV value of a cell represents the maximum elevation inside the respective area of the world.

Quality value (QTY):

The quality value of a cell contains information about the vertical and horizontal accuracy of the elevation data.

2.1 Reference Frames

Longitude-latitude positions are geographically referenced to the WGS-84 ellipsoid (see Figure 2-1).

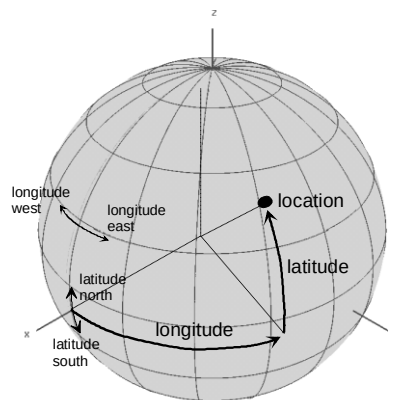


Figure 2-1: WGS-84 Ellipsoid

Cell locations are given in cell-centred format. Each post is centred inside the cell represented by the post. Therefore, the post represents an area with an extend of half the longitudinal resolution to the east and west and half of the latitudinal resolution to the north and south from its location

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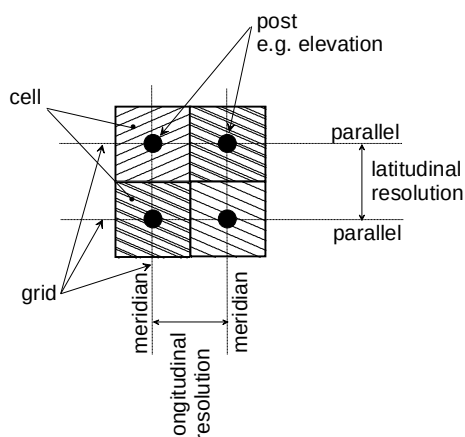


Figure 2-2: Cell Registration

Elevation values (ELV) are given with respect to the Mean Sea Level (MSL) using EGM96 projection.

2.2 Water Offset

In order to identify water areas, an offset value is added to all elevations values which represent sea or lakes. This allows both to identify water areas and to extract the elevation of the post by subtracting the offset. The offsets are negative numbers in order to avoid misinterpretation.

There are two water offsets:

- (Sea) water offset: -20000 Used for oceans and uncategorized water.
- Body water offset: -30000 Used for lakes and rivers.

Example:

A post -20000 means ocean with elevation 0m.

A post -29990 means lake or river with elevation 10m.

Note:

In cases where the body water information is not available, only the sea water offset is used (and becomes the “water offset”).

2.3 Segmentation

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The complete world model requires an enormous amount of memory. An appropriate way to structure the data is to divide the world into individual segments. In the Thales model, a segment is defined by a 1° by 1° area. The number of cells per segment depends on the resolution, e.g. a segment file with a resolution of 30" holds 120 by 120 cell values.

2.4 File Structure

The information belonging to a segment are not stored in a single file, but in multiple files with individual semantics. Thus in the project on hand, two files exist for each segment containing elevation data and quality values. Figure 2-3 shows this idea in general.

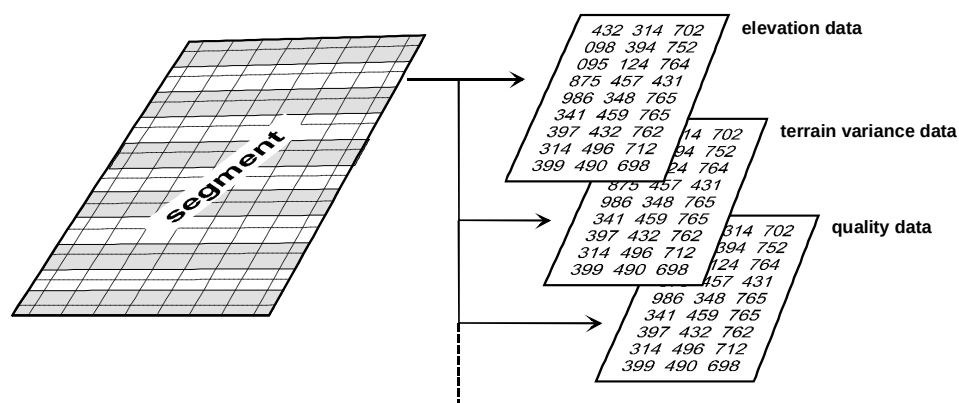


Figure 2-3: Segment Content

The grid-based data are stored in a row-major format. The enumeration starts in the north-west corner and ends in the south-east corner. Each row (or line) in a data set represents a constant latitude while the longitude ranges from the west to the east corner of the segment (see Figure 2-4).

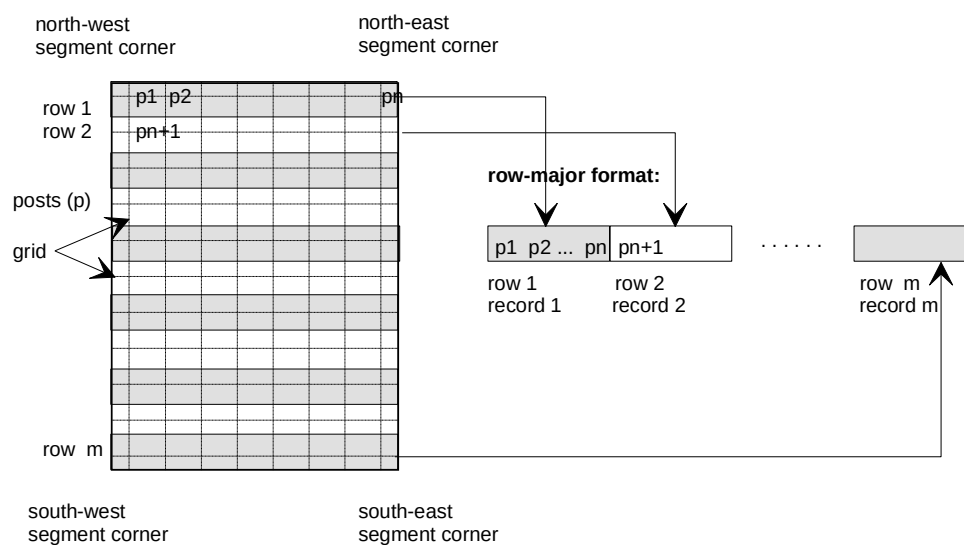


Figure 2-4: Row-Major Format

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3 Database Organization

Each database is delivered on a web server or a hard drive. Each delivery has the same directory structure as shown in Figure 3-1.

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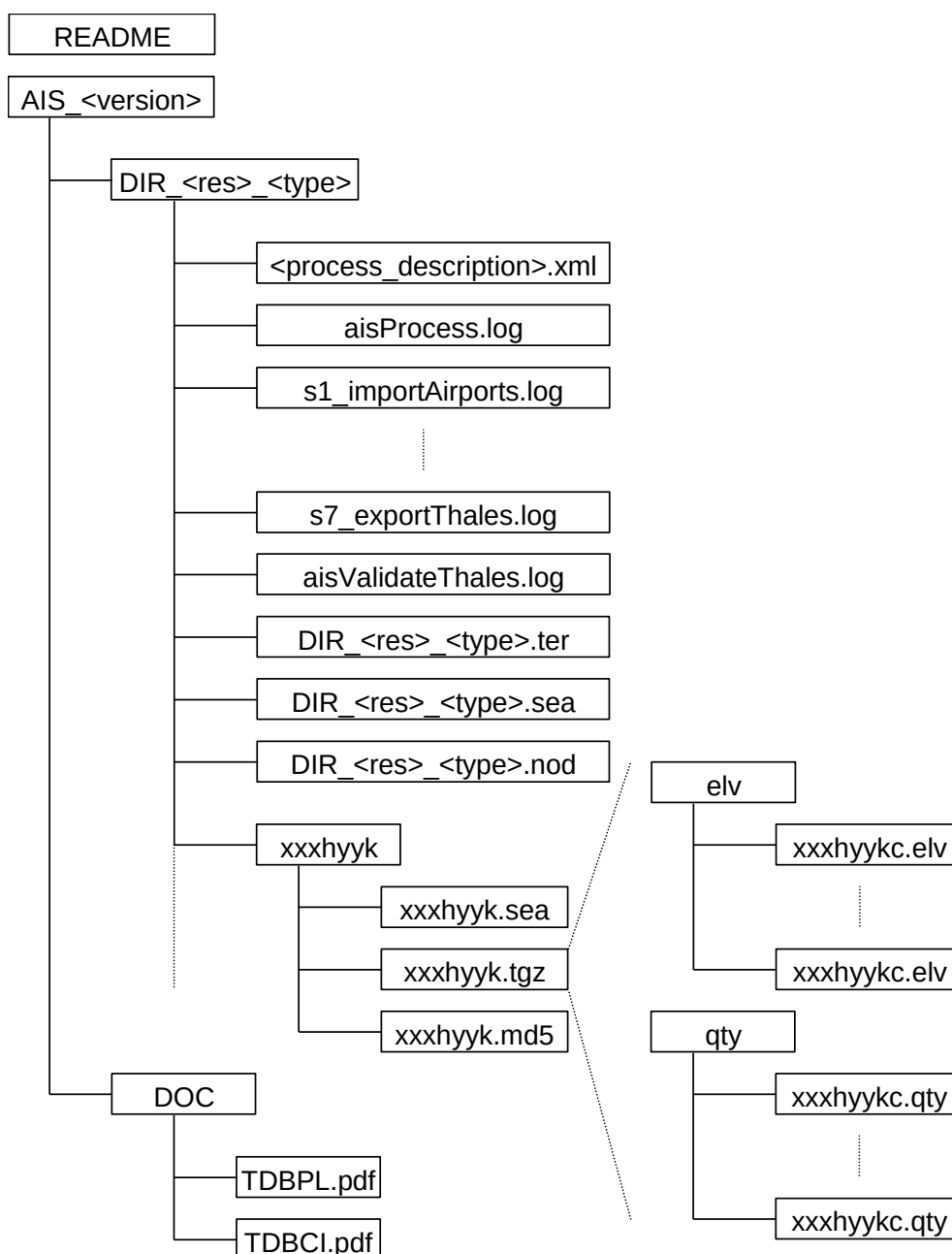


Figure 3-1: Directory structure on the database delivery.

The root directory of the delivery contains:

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- a README file describing the delivery contents,
- the main directory AIS_<version>. Here <version> is the database release identifier, e.g. "300".

Below the main directory, there are two directories:

- DOC, containing the automatically generated documents [TDBCI] and [TDBPL].
- DIR_<res>_<type>, the data directory with <res> in {15,30,3} and <type> in {MAX,AVG} containing four kinds of files:
 1. Process description file
 2. Log files
 3. Database information files
 4. Data directories

The next sections describe these files in detail.

3.1 Process Description File

The syntax of the process description file is described in [TDBPPD] in detail.

3.2 Log Files

The syntax of the process log file is described in [TDBPPD] in detail.

3.3 Database Information Files

These ASCII files contain information about the presence and content of the 1° by 1° segment files in the data base. They cover the whole world and containing 180 lines of 360 entries, each. Each entry represents a 1°x1° segment. The first line represents 90° north, the first column represents 180 west.

The following database information maps are provided:

- terrain segments (*.ter)
- sea segments (*.sea)
- no-data segments (*.nod)

An entry can be either 1 or 0 indicating the presence (or absence) of the respective data.
Entries are divided by spaces. An entry 1 means:

- In *.ter files: the presence of the respective segment file with terrain data
- In *.sea files: the absence of the respective segment file since all data would be sea level data.
- In *.nod files: the absence of the respective segment file since no data is available.

Note: 1) A consistency check can be performed, since the files must sum up to 1 at each position.

2) The case when a segment consists of only sea and NO_DATA values is solved by regularly storing the segment and indicating this by a "1" in the information file for terrain segments.

In addition to the entry block, the files can start with an arbitrary number of comment lines.
Each comment line begins with two slashes (/ /).

See a part of an exemplary *.ter file:

```
// world_30_max.ter
// Terrain segment overview file
// 10.10.2001
// (C) THALES Avionics - Digital Elevation File
1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
.
.
.
```

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3.4 Data Directories

The data directories are a set of 2592 directories corresponding to 5°x 5° areas, each of them being named `xxxhyyk` with

- `xxx` : longitude of the North-West corner
- `h` : e or w, indicating east or west
- `yy` : latitude of the North-West corner
- `k` : n or s, indicating north or south

Each directory contains three files:

- `xxxhyyk.tgz` : a gzipped tar file containing the following directory tree :
 - `elv/`
This directory contains all the `.elv` files of the tile. The naming of the files is described below.
 - `qty/`
This directory contains all the `.qty` files of the tile. The naming of the files is described below.
- `xxxhyyk.sea` : contains the list of segments (within the 5° x 5° tile) that are not provided because they contain only sea (1 per line, identified by their basename),
- `xxxhyyk.md5` : contains the CRC checksum of the files `xxxhyyk.tgz` and `xxxhyyk.sea` according to the MD5 checksum utility convention.

The segment files (elevation and quality segments) are named according to the following conventions:

```
xxxhyykc.eee
xxx: longitude of the western segment boundary in degrees
      (xxx in [001..180] in the west, xxx in [000..179] in the east)
h:   hemisphere code „w“ or „e“
yy:  latitude of the northern segment boundary in degrees
      (yy in [01..90] in the north, yy in [00..89] in the south)
k:   hemisphere code „n“ or „s“
c:   segment code „s“ for standard segments. This is used for future
      extensions.

eee: .elv indicates an elevation data file
     .qty indicates a quality data file
```

Example: The file `011e48ns.elv` represents the segment with the north-west corner at 11° east and 48° north which is an area in Austria (Europe). The segment is standard segment and contains elevation data.

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See Section 4 for a detailed file format description.

4 Segment File Format

4.1 Overview

In den project on hand, two different segment file types are used:

- Elevation files
- Quality files

Each segment file contains the following data blocks:

- General header information
- File specific header information
- File specific data records

The segment files are binary data files. Byte order is in Intel byte order, like PC's or DEC Alpha, which stores the least significant byte first ("little endian", LSB). On systems which use the Motorola byte order which stores the most significant byte first (big endian, MSB), the user have to swap bytes of the data. This is used by common UNIX workstations like Silicon Graphics or Sun SPARC.

A byte order identifier is provided to determine the byte order of the database. To check if the byte order of the current computer is identical to the byte order used within the database, a two byte integer which contains a pre-defined value is provided in the header block. This enables an easy and unambiguous test.

Within the following chapters an overview about the content of the segment files is given. Please refer to the reference chapter to get more detailed information about a specific data element.

4.2 General Header

The general header is included in all types of segment files (elevation and quality). Except for the file type identifier, it is identical throughout all files of a specific segment.

Byte Position	Description	Size (Bytes)	Ref.
0	Data type identifier	2	5.1
2	Byte order identifier	2	5.2
4	Byte order check value	2	5.3
6	Segment version code	4	5.4
10	Segment creation date	4	5.5
14	Segment name	14	5.6
28	Location reference	2	5.7
30	Elevation reference	2	5.8
32	Longitude of the NW corner	4	5.9
36	Latitude of the NW corner	4	5.10
40	Longitude of the NE corner	4	5.9
44	Latitude of the NE corner	4	5.10
48	Longitude of the SW corner	4	5.9
52	Latitude of the SW corner	4	5.10
56	Longitude of the SE corner	4	5.9
60	Latitude of the SE corner	4	5.10
64	Longitudinal segment size	4	5.11
68	Latitudinal segment size	4	5.12
72	Longitudinal resolution	4	5.13
76	Latitudinal resolution	4	5.14
80	Longitude of first post	4	5.15
84	Latitude of first post	4	5.16
88	Number of columns	4	5.17
92	Number of rows	4	5.18
96	Content identifier 1	2	5.19
98	Content identifier 2	2	5.19
100	Content identifier 3	2	5.19
102	Content identifier 4	2	5.19
104	Content identifier 5	2	5.19
106	Content identifier 6	2	5.19
108	Content identifier 7	2	5.19
110	Content identifier 8	2	5.19
112	Content identifier 9	2	5.19
114	Content identifier 10	2	5.19

The total size of the general header is 116 bytes.

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4.3 Elevation Files

The elevation files contain the actual elevation information.

4.3.1 File Structure

An elevation file consists of the following information blocks:

- general header
- elevation file header
- elevation data

The elevation data is divided into number_of_rows elevation records, each one representing elevation values of a constant latitude. Each elevation record starts with statistical information followed by number_of_columns elevation posts. The values number_of_rows and number_of_columns are defined in the general header.

4.3.2 Elevation File Header

Position	Description	Size (Bytes)	Ref.
0	Elevation file version code	4	5.4
4	Elevation file creation date	4	5.5
8	Elevation scale	2	5.20
10	Elevation type	2	5.21
12	Bytes per elevation (=2)	2	5.22
14	Minimal segment elevation	2	5.23
16	Maximal segment elevation	2	5.24
18	No data value (= -9999)	2	5.25

The total size of the elevation header is 20 bytes.

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4.3.3 Elevation Record:

Position	Description	Size (Bytes)	Ref.
0	Counter	4	5.26
4	Minimal record elevation	2	5.27
6	Maximal record elevation	2	5.28
8	Longitude first elevation post	4	5.29
12	Latitude first elevation post	4	5.30
16	First elevation post in this record	2	5.31
18	Second elevation post in this record	2	5.31
...		2	5.31
...	Last elevation post in this record	2	5.31

The size in bytes of each elevation record is

$$record_size = 14 + 2 * number_of_columns$$

and the total size in bytes of the elevation data block is

$$elevation_data_size = number_of_rows * record_size$$

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4.4 Quality Files

The quality files contain information about the horizontal and vertical quality of the elevation data.

4.4.1 File Structure

A quality file consists of the following information blocks:

- general header
- quality file header
- quality definition record(s)
- quality tag data

One or more quality definition records are provided, identified by a unique quality tag. The quality tag data assigns to each elevation post one of the defined quality definitions.

Note: In the project on hand, the quality files contain 7 quality definition records, each.

Like the elevation data, the quality tag data is divided into number_of_rows quality tag records, each one representing quality values of a constant latitude. Each record starts with statistical information followed by numer_of_columns tag identifications.

4.4.2 Quality File Header

Position	Description	Size (Bytes)	Ref.
0	Quality file version code	4	5.4
4	Quality file creation date	4	5.5
8	Accuracy scale	2	5.20
10	Number of quality definition records	2	5.32
12	Bytes per quality tag identification (=2)	2	5.22
14	Minimal segment quality tag identification	2	5.23
16	Maximal segment quality tag identification	2	5.24
18	No data value (=9999)	2	5.25

The total size of the quality header is 20 bytes.

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4.4.3 Quality Definition Record

Position	Description	Size (Bytes)	Ref.
0	Quality tag identification	2	5.33
2	Absolute horizontal accuracy - sigma 1	4	5.34
6	Relative horizontal accuracy - sigma 1	4	5.35
10	Absolute vertical accuracy - sigma 1	4	5.36
14	Relative vertical accuracy - sigma 1	4	5.37
18	Absolute horizontal accuracy - sigma 2	4	5.34
22	Relative horizontal accuracy - sigma 2	4	5.35
26	Absolute vertical accuracy - sigma 2	4	5.36
30	Relative vertical accuracy - sigma 2	4	5.37
34	Absolute horizontal accuracy - sigma 3	4	5.34
38	Relative horizontal accuracy - sigma 3	4	5.35
42	Absolute vertical accuracy - sigma 3	4	5.36
46	Relative vertical accuracy - sigma 3	4	5.37

The total size of each quality definition record is 50 bytes.

4.4.4 Quality Tag Record

Position	Description	Size (Bytes)	Ref.
0	Counter	4	5.26
4	Minimal record quality tag identification	2	5.27
6	Maximal record quality tag identification	2	5.28
8	Longitude first quality tag identification	4	5.29
12	Latitude first quality tag identification	4	5.30
16	First quality tag identification in this record	2	5.33
18	Second quality tag identification in this record	2	5.33
...		2	5.33
...	Last quality tag identification in this record	2	5.33

The size in bytes of each quality tag record is

$$record_size = 16 + 2 * number_of_columns$$

and the total size in bytes of the quality tag data block is

$$quality_data_size = number_of_rows * record_size$$

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5 Data Definition Reference

5.1 File Type Identifier

Type	integer
Size (Bytes)	2
Description	The file identifier identifies the type of information the file contains. Supported types of files are: 1 = Elevation 2 = Terrain variance 3 = Quality 4 = Trust value

5.2 Byte Order Identifier

Type	integer
Size (Bytes)	2
Description	The byte order identifier identifies the byte order used for the database. Supported byte orders are: 0 = unknown (not used in the binary databases delivered by Thales) 1 = LSB, little endian, (e.g. PC's, DEC Alpha) 2 = MSB, big endian, Motorola (e.g. Sun SPARC, SGI)

5.3 Byte Order Check Value

Type	integer
Size (Bytes)	2
Description	The byte order check value is a fixed integer which helps to ensure that the byte order of the target machine is identical to the one of the source computer. The value of this field is always set to: 24575

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5.4 Version Code

Type	integer
Size (Bytes)	4
Description	The version code contains the current version of the database.
Example	300 (for database with ID DEM_WORLD_30_MAX_300)

5.5 Creation Date

Type	integer
Size (Bytes)	4
Description	The creation date contains the date at which the file was created or revised. The format is day month year: DDMMYY
Example	250696 stands for June, 25, 1996

5.6 Segment Name

Type	string
Size (Bytes)	14
Description	The segment name is a string which contains the name of the segment in upper case and without the segment code component ("s"). The string is left justified. Empty characters are filled with spaces. (Note: The string does not end with a 0 value!)
Example	"011E48N " "00660E2880N "

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5.7 Location Reference

Type	integer
Size (Bytes)	2
Description	The location reference (LR) contains a value which defines the reference datum used for the location (longitude and latitude) of the posts. Supported datum are: LR = 1: WGS-84 ellipsoid

5.8 Elevation Reference

Type	integer
Size (Bytes)	2
Description	The elevation reference (ER) contains a value which define the reference datum used for the elevations and related data. Supported datum are: ER = 1: WGS-84 ellipsoid ER = 2: Mean Sea Level

5.9 Longitude Of Corner

Type	integer
Size (Bytes)	4
Description	The longitude of corner contains the geographic longitude of a corner of the area covered by the segment file. The longitude is expressed in arc-seconds. Positive values refer to longitude east, negative values refer to longitude west.
Example	39600 (equal to 11° East) -39600 (equal to 11° West)

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5.10 Latitude Of Corner

Type	integer
Size (Bytes)	4
Description	The latitude of corner contains the geographic latitude of a corner of the area covered by the segment file. The latitude is expressed in arc-seconds. Positive values refer to latitude north, negative values refer to latitude south.
Example	172800 (equal to 48° North) -172800 (equal to 48° South)

5.11 Longitudinal Segment Size

Type	integer
Size (Bytes)	4
Description	The longitudinal segment size is the size of the area covered by the segment file in longitude direction. The size is expressed in arc-seconds. It must be identical to the difference between the north-east and the north-west corner or the difference between the south-east and the south-west corner.
Example	10800 (equal to a 3° segment)

5.12 Latitudinal Segment Size

Type	integer
Size (Bytes)	4
Description	The latitudinal segment size is the size of the area covered by the segment file in latitude direction. The size is expressed in arc-seconds. It must be identical to the difference between the north-east and the south-east corner or the difference between the north-west and the south-west corner.
Example	10800 (equal to a 3° segment)

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5.13 Longitudinal Resolution

Type	integer
Size (Bytes)	4
Description	The longitudinal resolution defines the geographic distance between two columns (meridians) next to each other within the data set. The resolution is expressed in 1/100 arc-seconds.
Example	1500 (15 arc-seconds resolution)

5.14 Latitudinal Resolution

Type	integer
Size (Bytes)	4
Description	The latitudinal resolution defines the geographic distance between two rows (parallels) next to each other within the data set. The resolution is expressed in 1/100 arc-seconds.
Example	1500 (15 arc-seconds resolution)

5.15 Longitude Of First Post

Type	integer
Size (Bytes)	4
Description	This field contains the geographic longitude of the first post (the north-western post) within the data set. The longitude is expressed in 1/100 arc-seconds and indicates the center of the cell.
Example	3960750 (western boundary of the segment 11° east, longitudinal resolution 15.0 arc-seconds) -3959250 (western boundary of the segment 11° west, longitudinal resolution 15.0 arc-seconds)

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5.16 Latitude Of First Post

Type	Integer
Size (Bytes)	4
Description	This field contains the geographic latitude of the first post (the north-western post) within the data set. The latitude is expressed in 1/100 arc-seconds and indicates the center of the cell.
Example	17279250 (northern boundary of the segment 48° north, latitudinal resolution 15.0 arc-seconds) -17280750 (northern boundary of the segment 48° south, latitudinal resolution 15.0 arc-seconds)

5.17 Number Of Columns

Type	integer
Size (Bytes)	4
Description	The number of columns contains the number of columns (meridians) within the data set.
Example	120 (1° segment, longitudinal resolution 30 arc-seconds)

5.18 Number Of Rows

Type	integer
Size (Bytes)	4
Description	The number of rows contains the number of rows (parallels) within the data set.
Example	120 (1° segment, latitudinal resolution 30 arc-seconds)

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5.19 Content Identifier

Type	integer
Size (Bytes)	2
Description	The content identifier (CI) indicates which information are available for this segment file. A value of 0 indicates that the accompanied information is not available. A value of 1 indicates that the information is available. The following identifier are supported: Content identifier 1 : elevation data (shall always be set to 1) Content identifier 2 : terrain variance data Content identifier 3 : quality data Content identifier 4 : trust value Content identifier 5 : for further use Content identifier 6 : for further use Content identifier 7 : for further use Content identifier 8 : for further use Content identifier 9 : for further use Content identifier 10: for further use
Example	CI1 = 1; CI2 = 0; CI3 = 1; CI4 = 0 indicates a segment which contains elevation and quality data.

5.20 Data Scale

Type	integer
Size (Bytes)	2
Description	The data scale contains a value with which the data in the following data block has to be divided to achieve a value expressed in meters.
Example	1 -> data are expressed in meter 100 -> data are expressed in centimeter

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5.21 Elevation Type

Type	integer
Size (Bytes)	2
Description	The elevation type (ET) contains a value which defines the type of the elevation. Supported types are: ET = 0: the type of elevation is unknown ET = 1: elevation values express the minimal terrain elevations which can occur within their cells ET = 2: elevation values express the maximal terrain elevations which can occur within their cells ET = 3: elevation values express the mean value of the terrain elevations which can occur within their cells ET = 4: elevation values express a weighted average of the terrain elevations which can occur within their cells

5.22 Bytes Per Post

Type	integer
Size (Bytes)	2
Description	The bytes per post contains a value which defines the number of bytes which are used for the following data within the current file.
Example	2 (the size of the following data of the current file is two byte)

5.23 Minimal Segment Post

Type	Integer
Size (Bytes)	bytes per post
Description	The minimal segment post contains the minimal post value of the current data within the whole segment. <i>Note: Refer to the data scale field of the current file to determine which unit is used for the minimal segment post.</i>

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5.24 Maximal Segment Post

Type	Integer
Size (Bytes)	bytes per post
Description	The maximal segment post contains the maximal post value of the current data within the whole segment. <i>Note: Refer to the data scale field of the current file to determine which unit is used for the maximal segment post.</i>

5.25 No Data Value

Type	integer
Size (Bytes)	bytes per post of the current file
Description	The NO_DATA value contains a value which indicates that no information is available for the current cell.
Example	-9999 is a typical NO_DATA value for two byte terrain data

5.26 Record Counter

Type	integer
Size (Bytes)	4
Description	The record counter contains a value which identifies the number of the current record. The range of the counter is between 0 (most northern row within the segment) and the number of rows minus 1 (most southern row within the segment)

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5.27 Minimal Record Post

Type	integer
Size (Bytes)	bytes per post
Description	The minimal record post contains the minimal post value within the current record. NO_DATA posts are not considered for the minimum computation. Only if all posts in the record are NO_DATA, the minimum value is set to NO_DATA. <i>Note: refer to the data scale field of the current file to determine which unit is used for the minimal record post.</i>

5.28 Maximal Record Post

Type	integer
Size (Bytes)	bytes per post
Description	The maximal record post contains the maximal post value within the current record. If all posts in the record are NO_DATA, the minimum value is set to NO_DATA. <i>Note: refer to the data scale field of the current file to determine which unit is used for the maximal record post.</i>

5.29 Longitude First Record Post

Type	integer
Size (Bytes)	4
Description	This field contains the geographic longitude of the first post (the western post) within the current record. The longitude is expressed in 1/100 arc-seconds and indicates the center of the post.
Example	3960750 (western boundary of the segment 11° east, longitudinal resolution 15.0 arc-seconds) -3959250 (western boundary of the segment 11° west, longitudinal resolution 15.0 arc-seconds)

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5.30 Latitude First Record Post

Type	Integer
Size (Bytes)	4
Description	This field contains the geographic latitude of the first post (the western post) within the current record. The latitude is expressed in 1/100 arc-seconds and indicates the center of the post.
Example	17279250 (northern boundary of the segment 48° north, latitudinal resolution 15.0 arc-seconds first record) -17280750 (northern boundary of the segment 48° south, latitudinal resolution 15.0 arc-seconds first record)

5.31 Elevation Post

Type	integer
Size (Bytes)	bytes per elevation
Description	The elevation post contains the elevation for the defined cell. <i>Note: refer to the data scale field of the current file to determine which unit is used for the elevation post.</i>

5.32 Number Of Quality Definitions

Type	integer
Size (Bytes)	2
Description	The number of quality definitions contains a value which defines, how much quality definition records are present within this file.

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5.33 Quality Tag Identification

Type	integer
Size (Bytes)	bytes per quality tag
Description	The quality tag identification contains a unique value for each quality definition. This tag is used in the quality tag records and assigns a quality as defined in the associated quality definition to each elevation post. The range is from 0 to the number of quality definitions.

5.34 Absolute Horizontal Accuracy

Type	integer
Size (Bytes)	4
Description	The absolute horizontal accuracy contains a value which specifies the uncertainty of the absolute horizontal location of a given location as a circular error. The following confidence levels are supported: sigma 1: = 68.26% confidence level sigma 2: = 95.44% confidence level sigma 3: = 99.73% confidence level <i>Note: refer to the data scale field of the current file to determine which unit is used for the accuracy.</i> <i>Note: The NO_DATA value indicates that no value is available.</i>

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5.35 Relative Horizontal Accuracy

Type	Integer
Size (Bytes)	4
Description	The relative horizontal accuracy contains a value which specifies the uncertainty of a horizontal distance between two given location as a circular error. The following confidence levels are supported: sigma 1: = 68.26% confidence level sigma 2: = 95.44% confidence level sigma 3: = 99.73% confidence level <i>Note: refer to the data scale field of the current file to determine which unit is used for the accuracy.</i> <i>Note: The NO_DATA value indicates that no value is available.</i>

5.36 Absolute Vertical Accuracy

Type	integer
Size (Bytes)	4
Description	The absolute vertical accuracy contains a value which specifies the uncertainty of the elevation of a given elevation. The following confidence levels are supported: sigma 1: = 68.26% confidence level sigma 2: = 95.44% confidence level sigma 3: = 99.73% confidence level <i>Note: refer to the data scale field of the current file to determine which unit is used for the accuracy.</i> <i>Note: The NO_DATA value indicates that no value is available.</i>

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5.37 Relative Vertical Accuracy

Type	integer
Size (Bytes)	4
Description	The relative vertical accuracy contains a value which specifies the uncertainty of the elevation between two given elevation posts. The following confidence levels are supported: sigma 1: = 68.26% confidence level sigma 2: = 95.44% confidence level sigma 3: = 99.73% confidence level <i>Note: refer to the data scale field of the current file to determine which unit is used for the accuracy.</i> <i>Note: The NO_DATA value indicates that no value is available.</i>

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6 Referenced Documents

6.1 Applicable documents

[DO-200A]	RTCA / DO-200A - EUROCAE / ED-76 – Standards for Processing Aeronautical Data
[DO-200B]	RTCA / DO-200B - EUROCAE / ED-76A – Standards for Processing Aeronautical Data

6.2 Project documents

6.2.1 Database Tools

[TDBTR]	Terrain Data Base Tools Requirements - J43059
[TDBTS]	Terrain Data Base Tools Specification - J43060
[TDBTCI]	Terrain Data Base Tools Configuration Index - J43066

6.2.2 Database Production

[TDBDR]	Terrain Data Base Data Requirements - J43054
[TDBFDD]	Terrain Data Base Format Description Document - J43055
[TDBPPD]	Terrain Database Production Process Description – J43057

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